EXP NO: 20 **Experiment to understand K-means clustering algorithm for a given data set.**

AIM:

To understand and implement the K-means clustering algorithm on a given dataset to group similar data points into clusters based on their features.

PROGRAM:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.cluster import KMeans

from google.colab import files


uploaded = files.upload()

df = pd.read_csv('Mall_Customers.csv')


df.info()

print(df.head())


sns.pairplot(df)

plt.show()


features = df.iloc[:, [3, 4]].values


model = KMeans(n_clusters=5, random_state=42, n_init=10)

model.fit(features)


Final = df.iloc[:, [3, 4]].copy()
```

```
Final['label'] = model.predict(features)


print(Final.head())


sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8) \
    .map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
    .add_legend()


plt.show()


features_el = df.iloc[:, [2, 3, 4]].values
wcss = []


for i in range(1, 10):
    model = KMeans(n_clusters=i, random_state=42, n_init=10)
    model.fit(features_el)
    wcss.append(model.inertia_)


plt.plot(range(1, 10), wcss, marker='o')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.title('Elbow Method')
plt.show()
```

OUTPUT:

```
#   Column                Non-Null Count  Dtype
---  ------                --------------  -----
```

0 CustomerID 200 non-null int64

1 Gender 200 non-null object

2 Age 200 non-null int64

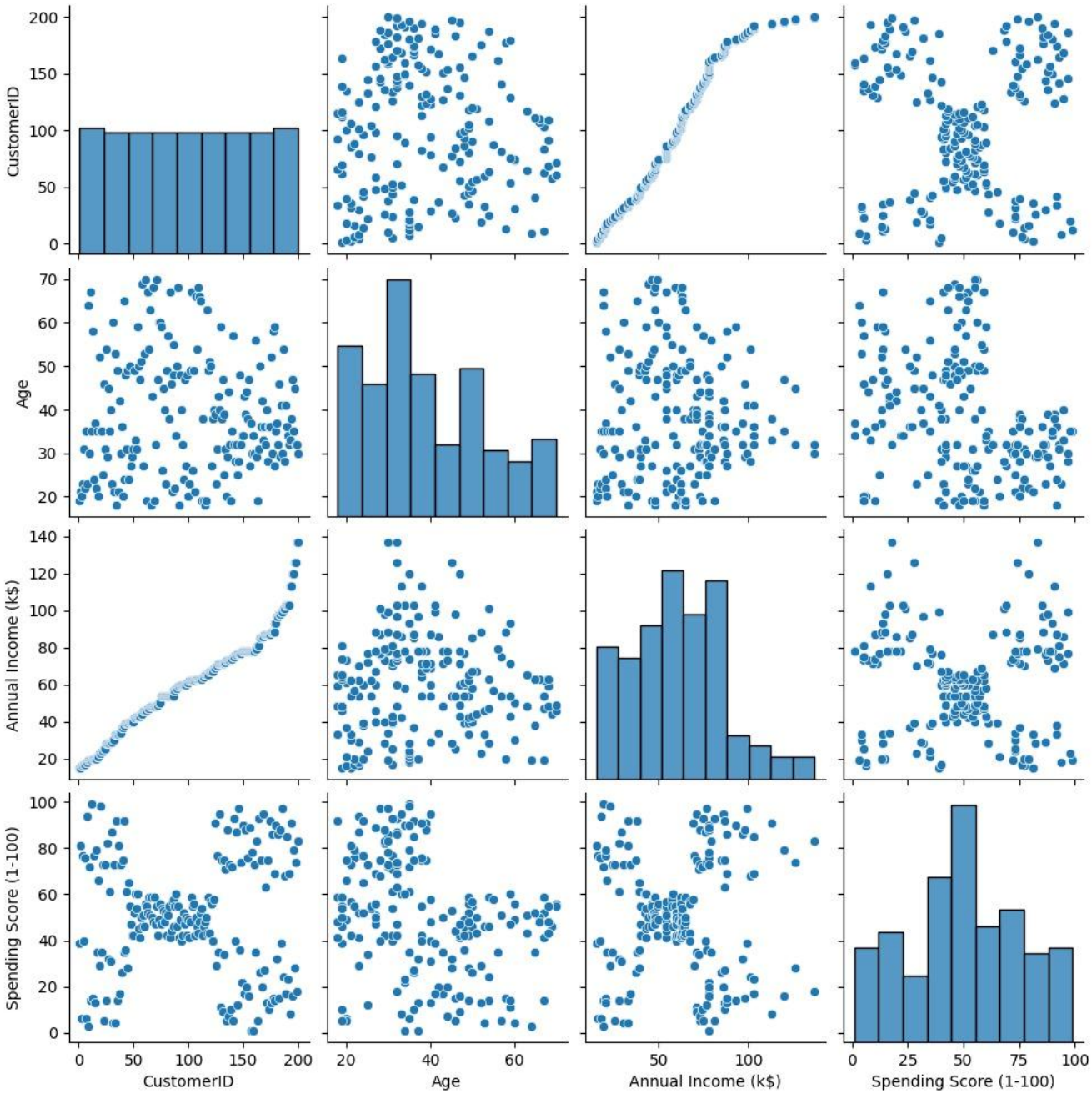
3 Annual Income (k$) 200 non-null int64

4 Spending Score (1-100) 200 non-null int64

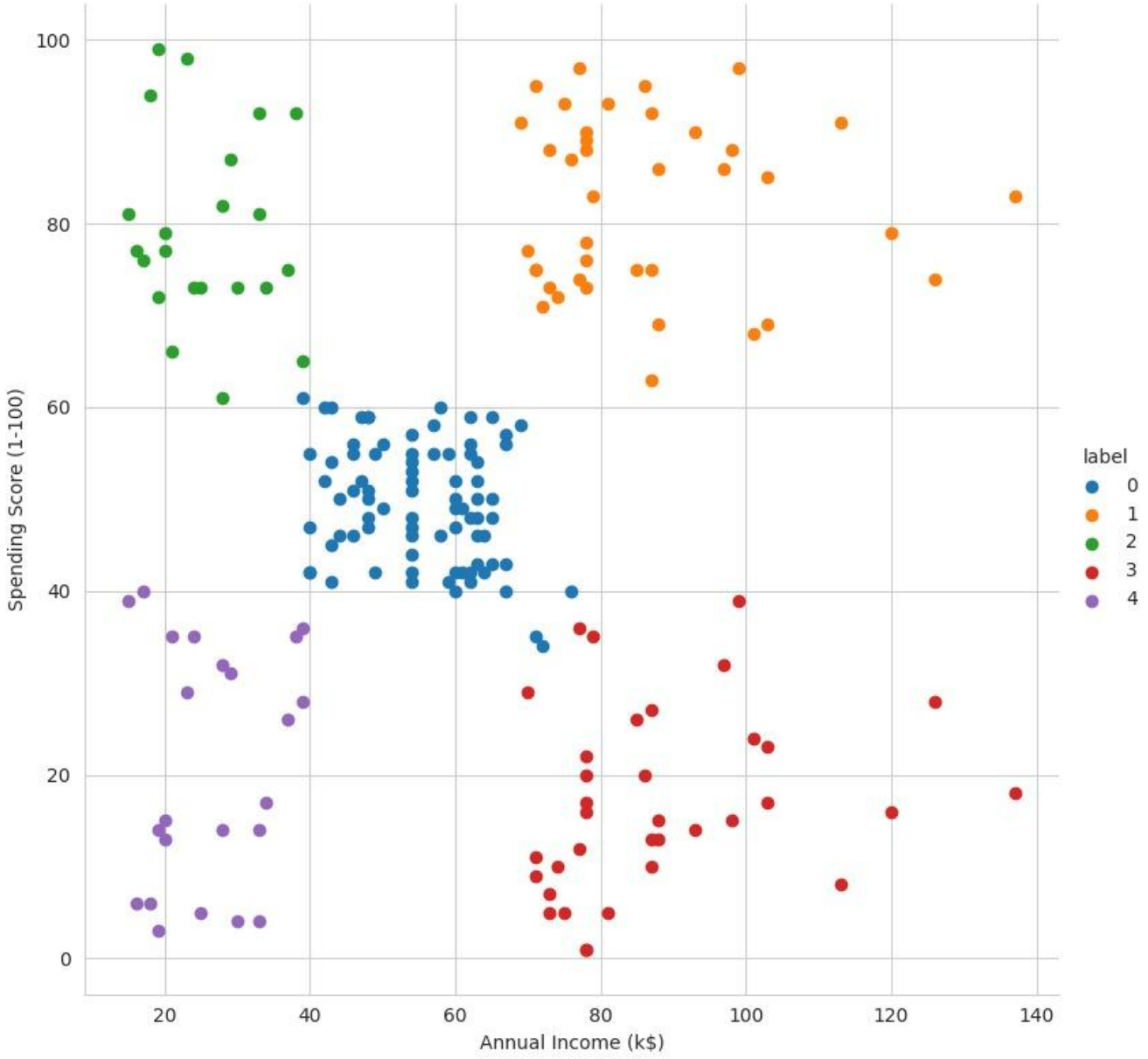
dtypes: int64(4), object(1)

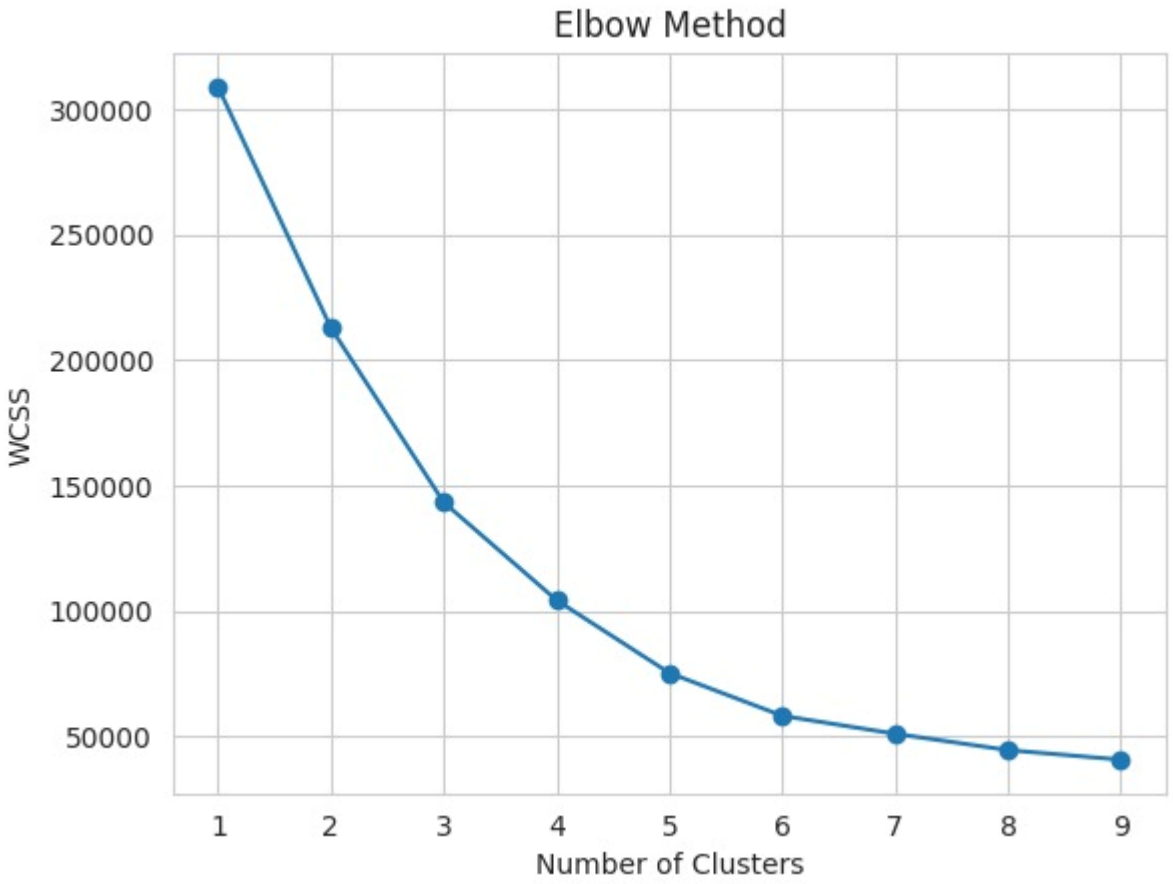
memory usage: 7.9+ KB

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40



	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4


Spending Score (1-100)
Annual Income (k$)
label
0
1
2
3
4
0
20
40
60
80
100
20
40
60
80
100
120
140



RESULT:

Thus, the K-means clustering algorithm was successfully implemented on the given dataset, and the data points were grouped into clusters based on their similarities.